

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0709

Okawville 138 kV
October 19, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0709. The Interconnection Customer proposes a 10.4 MW battery energy storage facility interconnecting at the Okawville 138 kV substation in the Pennsylvania area. The request was received on November 29, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 349 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 42 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$156,507,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of WALNUT – W GALLATIN 345kV L, the BULLMOOSE – MOROVIS 8801 138kV loads to 232.8% of its 248 MVA rating. Project DFAX: 33.8%.

For the loss of WALNUT – W GALLATIN 345kV, the MOROVIS 801 – NEWCOMERSTOWN 138kV loads to 205.6% of its 289 MVA rating. Project DFAX: 33.8%.

DORNYVILLE – WESCOSVILLE 138kV: 192.1% loading (213 MVA) following the outage of the MAGIC KINGDOM – OKAWVILLE 138kV; DFAX = 43.9%.

For the loss of WALNUT – WEST GALLATIN 345kV, the NEWCOMERSTOWN – VALLEY HOME TAP 2 138kV loads to 187.6% of its 341 MVA rating. Project DFAX: 33.8%.

The W GALLATIN – MEREDOSIA 138kV is overloaded to 186.2% of its 261 MVA emergency rating for the loss of the WEST GALLATIN – FELIDA 345kV. The project contributes 12.3% DFAX (1.3 MW) to this overload.

The E ARLINGTON – VLY HOME TAP 2 138kV is overloaded to 181.9% of its 362 MVA emergency rating for the loss of the W GALLATIN – WALNUT 345kV. The project contributes 33.8% DFAX (3.5 MW) to this overload.

The E ARLINGTON – WESCOSVILLE 138kV is overloaded to 180.1% of its 218 MVA emergency rating for the loss of the MAGIC KINGDOM – OKAWVILLE 138kV line. The project contributes 43.9% DFAX (4.6 MW) to this overload.

DORNYVILLE – EDWARDSPORT STATION 138kV: 172.0% loading (251 MVA) following the outage of the MAGIC KINGDOM – OKAWVILLE 138kV L; DFAX = 43.9%.

For the loss of MOUNT OLIVE – EDWARDSPORT STA 138kV, the LANDIS DELIVERY NO 2 SUBSTATION – E ARLINGTON 138kV loads to 171.4% of its 225 MVA rating. Project DFAX: 44.6%.

For the loss of MOUNT OLIVE – EDWARDSPORT STA 138kV line, the LANDIS DELIVERY NO 2 – OKAWVILLE 138kV loads to 170.3% of its 232 MVA rating. Project DFAX: 44.6%.

MAGIC KINGDOM – OKAWVILLE 138kV Line: 168.7% loading (242 MVA) following the outage of the EDWARDSPORT STATION – MOUNT OLIVE 138kV; DFAX = 55.4%.

MOUNT OLIVE – EDWARDSPORT SA SUBSTATION 138kV Line: 166.8% loading (272 MVA) following the outage of the MAGIC KINGDOM – OKAWVILLE 138kV; DFAX = 43.9%.

For the loss of FELIDA – W GALLATIN 345kV, the SATSOP – MEREDOSIA 138kV loads to 162.9% of its 250 MVA rating. Project DFAX: 12.3%.

For the loss of FELIDA – TALENENERGY MARTINS CRK LLC HARWOOD 345kV, the FELIDA 500 kV/345 kV Transformer Bank loads to 155.1% of its 873 MVA rating. Project DFAX: 15.4%.

The FELIDA 345/138 kV TX is overloaded to 141.2% of its 560 MVA emergency rating for the loss of the FELIDA – W GALLATIN 345kV Line. The project contributes 16.8% DFAX (1.8 MW) to this overload.

The VANDALIA – MOUNT OLIVE SUBSTATION 138kV is overloaded to 137.5% of its 269 MVA emergency rating for the loss of the OKAWVILLE – MAGIC KINGDOM 138kV line. The project contributes 31.3% DFAX (3.3 MW) to this overload.

The ASH ST STATION 223 SUBSTATION – TOMBSTONE 345kV is overloaded to 136.4% of its 1,197 MVA emergency rating for the loss of the TOMBSTONE – MARCUS HOOK ENERGY LP 500kV. The project contributes 6.1% DFAX (0.6 MW) to this overload.

The VANDALIA SUBSTATION – KERHONKSON 138kV L is overloaded to 135.8% of its 263 MVA emergency rating for the loss of the PRINCETON MN – FELIDA 138kV. The project contributes 16.8% DFAX (1.7 MW) to this overload.

CLEAR SPRINGS – WALNUT 345kV Line: 134.5% loading (1,101 MVA) following the outage of the TOMBSTONE – BIXBY HILL 500kV; DFAX = 7.9%.

For the loss of FELIDA – W GALLATIN 345kV, the W ALLATIN 345/138 kV Transformer loads to 129.4% of its 375 MVA rating. Project DFAX: 12.3%.

For the loss of MONTOUR STA – ACADIA 500kV Line, the DREWS – ST NICHOLAS COGEN PROJECT 138kV Line loads to 127.0% of its 157 MVA rating. Project DFAX: 6.9%.

For the loss of KERHONKSON – VADALIA 138kV, the PRINCETON MN – FELIDA 138kV loads to 123.5% of its 279 MVA rating. Project DFAX: 16.7%.

For the loss of CORONET – WESTWOOD GENERATON LLC 500kV, the MONTOUR STATION 500-345 kV TX loads to 123.2% of its 1,067 MVA rating. Project DFAX: 8.6%.

IRVING SUBSTATION – MAGIC KINGDOM 138kV: 120.3% loading (234 MVA) following the outage of the W GALLATIN – FLIDA 345kV; DFAX = 21.0%.

FELIDA – TALENENERGY MARTINS CRK LLC HARWOOD 345kV: 117.0% loading (1,479 MVA) following the outage of the FELIDA 500/345 kV Transformer Bank; DFAX = 20.6%.

The CLEAR SPRINGS 500/345kV TX is overloaded to 117.0% of its 1,228 MVA emergency rating for the loss of the WIGWAM – UPSCO 345kV line. The project contributes 5.5% DFAX (0.6 MW) to this overload.

The JH WARDEN 500-345 kV Transformer Bank is overloaded to 115.7% of its 678 MVA emergency rating for the loss of the CORONET – WESTWOOD GENERATION LLC 500kV. The project contributes 5.5% DFAX (0.6 MW) to this overload.

For the loss of WALNUT SUBSTATION – CLEAR SPRINGS 345kV, the QUARRY – RIPARIA 138kV loads to 115.4% of its 191 MVA rating. Project DFAX: 8.5%.

For the loss of TOMBSTONE – BIXBY HILL 500kV, the W GALLATIN – FELIDA 345kV loads to 114.9% of its 1,554 MVA rating. Project DFAX: 8.9%.

The TALENENERGY MARTIN CRK LLC HARWOOD 500 kV/1 kV Transformer Bank 1 is overloaded to 111.9% of its 1,474 MVA emergency rating for the loss of the FELIDA 500-345 kV XFMR. The project contributes 15.9% DFAX (1.7 MW) to this overload.

For the loss of FELIDA 500/345kV Transformer Bank, the TALENENERGY MARTINS CRK LC HARWOOD 345/1 kV TX loads to 111.9% of its 1,474 MVA rating. Project DFAX: 15.9%.

The HICKORY WITHE – SIEGFRIED 138kV is overloaded to 110.2% of its 204 MVA emergency rating for the loss of the MOUNT OLIVE – VANDALIA 138kV. The project contributes 13.3% DFAX (1.4 MW) to this overload.

The REC LAPRAIRIE – SUMNER 138kV is overloaded to 109.9% of its 233 MVA emergency rating for the loss of the CLEAR SPRINGS – WALNUT 345kV. The project contributes 8.5% DFAX (0.9 MW) to this overload.

For the loss of VANDALIA – MOUNT OLIVE 138kV, the MONARCH – MARTINSILLE E 138kV loads to 109.5% of its 238 MVA rating. Project DFAX: 13.3%.

RIPARIA – REC LAPRAIRIE 138kV: 109.0% loading (223 MVA) following the outage of the CLEAR SPRINGS – WALNUT 345kV; DFAX = 8.5%.

SIEGFRIED – MARTINSVILLE E 138kV: 107.8% loading (229 MVA) following the outage of the MOUNT OLIVE – VANDALIA SUBSTATION 138kV Line; DFAX = 13.3%.

SATSOP – FELIDA 138kV: 106.5% loading (237 MVA) following the outage of the WEST GALLATIN SUBSTATION – FELIDA 345kV; DFAX = 6.8%.

The HICKORY WITHE – MOUNT OLIVE 138kV Line is overloaded to 106.1% of its 190 MVA emergency rating for the loss of the MOUNT OLIVE – VANDALIA 138kV. The project contributes 13.3% DFAX (1.4 MW) to this overload.

UPSCO – WIGWAM 345kV: 104.7% loading (1,213 MVA) following the outage of the WALNUT – W GALLATIN 345kV Line; DFAX = 6.8%.

The QUARRY – GEIGY 138kV is overloaded to 102.9% of its 195 MVA emergency rating for the loss of the CLEAR SPRINGS – WALNUT 345kV line. The project contributes 8.5% DFAX (0.9 MW) to this overload.

The WHEELABRATOR FRACKVILLE ENERGY – UPSCO 345kV is overloaded to 102.5% of its 1,225 MVA emergency rating for the loss of the WEST GALLATIN – WALNUT 345kV. The project contributes 6.0% DFAX (0.6 MW) to this overload.

BULLMOOSE – SUMNER 138kV: 101.9% loading (258 MVA) following the outage of the BULLMOOSE 345-1 kV Transformer Bank; DFAX = 20.5%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BULLMOOSE – MOROVIS 8801 138kV	Rebuild line with higher-capacity conductor	\$1,728,000	100% (\$1,728,000)
MOROVIS 801 – NEWCOMERSTOWN 138kV	Rebuild line with higher-capacity conductor	\$3,490,000	100% (\$3,490,000)
DORNYVILLE – WESCOSVILLE 138kV	Rebuild line with higher-capacity conductor	\$2,418,000	100% (\$2,418,000)
NEWCOMERSTOWN – VALLEY HOME TAP 2 138kV L	Rebuild line with higher-capacity conductor	\$4,877,000	100% (\$4,877,000)
W GALLATIN – MEREDOSIA 138kV Line	Rebuild line with higher-capacity conductor	\$2,283,000	100% (\$2,283,000)
EAST ARLINGTON – VALLEY HOME TAP 2 138kV	Rebuild line with higher-capacity conductor	\$1,918,000	100% (\$1,918,000)
WESCOSVILLE – E ARLINGTON 138kV	Rebuild line with higher-capacity conductor	\$3,166,000	100% (\$3,166,000)
DORNYVILLE – EDWARDSPORT STATION 138kV	Rebuild line with higher-capacity conductor	\$6,233,000	100% (\$6,233,000)
LANDIS DELIVERY NO 2 SUBSTATION – E ARLINGTON 138kV	Rebuild line with higher-capacity conductor	\$7,610,000	100% (\$7,610,000)
LANDIS DELIVERY NO 2 – OKWVILLE 138kV	Rebuild line with higher-capacity conductor	\$5,284,000	100% (\$5,284,000)
MAGIC KINGDOM – OKAWVILLE 138kV Line	Rebuild line with higher-capacity conductor	\$7,850,000	100% (\$7,850,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
EDWARDSPORT STATION – MOUNT OLIVE 138kV	Rebuild line with higher-capacity conductor	\$1,435,000	100% (\$1,435,000)
SATSOP – MEREDOSIA 138kV	Rebuild line with higher-capacity conductor	\$15,069,000	100% (\$15,069,000)
FELIDA 500/345kV Transformer Bank	Replace transformer with higher-rated unit	\$21,600,000	1% (\$174,000)
FELIDA 345/138 kV TX	Replace transformer with higher-rated unit	\$12,752,000	100% (\$12,752,000)
VANDALIA – MOUNT OLIVE SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$5,770,000	100% (\$5,770,000)
ASH ST STATON 223 – TMBSTONE 345kV	Rebuild line with higher-capacity conductor	\$25,078,000	0% (\$32,000)
VANDALIA – KERHONKSON 138kV line	Rebuild line with higher-capacity conductor	\$3,627,000	100% (\$3,627,000)
CLEAR SPRINGS – WALNUT 345kV	Rebuild line with higher-capacity conductor	\$41,949,000	0% (\$105,000)
W ALLATIN 345/138 kV Transformer	Replace transformer with higher-rated unit	\$18,861,000	100% (\$18,861,000)
DREWS – ST NICHOLAS COGEN PROJECT 138kV Line	Rebuild line with higher-capacity conductor	\$3,024,000	100% (\$3,024,000)
FELIDA – PRINCETON MN 138kV Line	Rebuild line with higher-capacity conductor	\$7,357,000	100% (\$7,357,000)
MONTOUR STA 500/345kV Auto	Replace transformer with higher-rated unit	\$45,051,000	0% (\$102,000)
IRVING SUBSTATION – MAGIC KINGDOM 138kV	Rebuild line with higher-capacity conductor	\$31,107,000	100% (\$31,107,000)
FELIDA – TALENENERGY MARTINS CRK LLC HARWOOD 345kV	Rebuild line with higher-capacity conductor	\$20,377,000	1% (\$132,000)
CLEAR SPRINGS 500 kV/345 kV Autotransformer	Replace transformer with higher-rated unit	\$43,877,000	0% (\$59,000)
JH WARDEN 500/345kV Auto	Replace transformer with higher-rated unit	\$38,062,000	0% (\$165,000)
QUARRY – RIPARIA 138kV L	Rebuild line with higher-capacity conductor	\$1,072,000	100% (\$1,072,000)
W GALLATIN – FELIDA 345kV	Rebuild line with higher-capacity conductor	\$29,886,000	0% (\$77,000)
TALENENERGY MARTINS CR LLC HARWOOD 500/1kV TX	Replace transformer with higher-rated unit	\$31,879,000	0% (\$159,000)
TALENENERGY MARTINS CRK LC HARWOOD 345/1 kV TX	Replace transformer with higher-rated unit	\$21,737,000	0% (\$108,000)
HICKORY WITHE – SIEGFRIED SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$1,086,000	100% (\$1,086,000)
REC LAPRAIRIE – SUMNER 138kV	Reconductor line	\$1,861,000	100% (\$1,861,000)
MONARCH – MARTINSILLE E 138kV	Reconductor line	\$807,000	100% (\$807,000)
RIPARIA – REC LAPRAIRIE 138kV	Reconductor line	\$753,000	100% (\$753,000)
SIEGFRIED – MARTINSVILLE E 138kV	Reconductor line	\$912,000	100% (\$912,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SATSOP – FELIDA 138kV Line	Reconductor line	\$952,000	100% (\$952,000)
HICKORY WITHE – MOUNT OLIVE 138kV Line	Reconductor line	\$989,000	100% (\$989,000)
UPSCO – WIGWAM 345kV	Reconductor line	\$19,390,000	0% (\$29,000)
QUARRY – GEIGY 138kV	Reconductor line	\$247,000	100% (\$247,000)
UPSCO – WHEELABRATOR FRACKVILLE ENERGY 345kV	Reconductor line	\$8,416,000	0% (\$14,000)
BULLMOOSE – SUMNER 138kV	Reconductor line	\$813,000	100% (\$813,000)
	Total allocated to project		\$156,507,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.